

Database Management Systems: Assignment-10

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Create the below given table and insert the data accordingly.

Create Table Job (job_id, job_title, min_sal, max_sal)

COLUMN NAME	DATA TYPE
job_id	Varchar2(15)
job_title	Varchar2(30)
min_sal	Number(7,2)
max_sal	Number(7,2)

Create table Employee (emp_no, emp_name, emp_sal, emp_comm, dept_no)

COLUMN NAME	DATA TYPE
emp_no	Number(3)
emp_name	Varchar2(30)
emp_sal	Number(8,2)
emp_comm	Number(6,1)
dept_no	Number(3)

Create table deposit(a_no,cname,bname,amount,a_date)

COLUMN NAME	DATA TYPE
a_no	Varchar2(5)
cname	Varchar2(15)
bname	Varchar2(10)
amount	Number(7,2)
a_date	Date

Create table borrow(loanno,cname,bname,amount).

COLUMN NAME	DATA TYPE
loanno	Varchar2(5)
cname	Varchar2(15)
bname	Varchar2(10)
amount	Varchar2(7,2)

Insert five records in the tables given above and perform followings:

		job_id varchar(15)	job_title varchar(30)	min_sal decimal(7,2)	max_sal decimal(7,2)
	>	J01	Manager	5000.00	12000.00
	>	J02	Clerk	2000.00	5000.00
	>	J03	Analyst	4000.00	10000.00
	>	J04	Salesman	3000.00	8000.00
	>	J05	HR	3500.00	7000.00

		emp_no decimal(3,0)	emp_name varchar(30)	emp_sal decimal(8,2)	emp_comm decimal(6,1)	dept_no decimal(3,0)	job_id varchar(15)	manager_id int(3)	hire_date date
	>	101	Anita	6000.00	500.0	10	J02	104	1995-05-20
	>	102	Anish	5500.00	(NULL)	20	J03	104	1996-07-15
	>	103	Ankit	5200.00	300.0	20	J04	104	1997-09-10
	>	104	Sunil	7000.00	800.0	30	J01	(NULL)	1998-01-05
	>	105	Manoj	6500.00	(NULL)	30	J05	104	1998-12-22

		a_no varchar(5)	cname varchar(15)	bname varchar(10)	amount decimal(7,2)	a_date date
	>	A01	Anil	Surat	2500.00	2006-01-10
	>	A02	Sunil	Mumbai	4000.00	2006-06-15
	>	A03	Pramod	Delhi	3000.00	2006-07-30
	>	A04	Ravi	Surat	2000.00	2006-02-20
	>	A05	Amit	Delhi	4500.00	2006-12-25

		loanno varchar(5)	cname varchar(15)	bname varchar(10)	amount decimal(7,2)
	>	L01	Anil	Surat	1500.00
	>	L02	Sunil	Mumbai	3000.00
	>	L03	Pramod	Delhi	2500.00
	>	L04	Ravi	Surat	3500.00
	>	L05	Amit	Delhi	4000.00

1. Give details of account no. and deposited rupees of customers having account opened between dates 01-01-06 and 25-07-06.

Query:

```
SELECT a_no, amount FROM deposit WHERE a_date BETWEEN '2006-01-01' AND '2006-07-25';
```

Output:

		a_no varchar(5)	amount decimal(7,2)
	>	A01	2500.00
	>	A02	4000.00
	>	A04	2000.00

2. Display name and salary of employee whose department no is 20. Give alias name to name of employee.

Query:

```
SELECT emp_name AS "Employee Name", emp_sal FROM Employee WHERE dept_no = 20;
```

Output:

		Employee Name	emp_sal decimal(8,2)
	>	Anish	5500.00
	>	Ankit	5200.00

3. Display name, number and salary of those employees whose name is 5 characters long and first three characters are 'Ani'.

Query:

```
SELECT emp_name, emp_no, emp_sal FROM Employee WHERE emp_name LIKE 'Ani__';
```

Output:

		emp_name varchar(30)	emp_no decimal(3,0)	emp_sal decimal(8,2)
	>	Anita	101	6000.00
	>	Anish	102	5500.00

4. Display the non-null values of employees and also employee name second character should be 'n' and string should be 5 character long.

Query:

```
SELECT * FROM Employee WHERE emp_name LIKE '_n___';
```

Output:

		emp_no decimal(3,0)	emp_name varchar(30)	emp_sal decimal(8,2)	emp_comm decimal(6,1)	dept_no decimal(3,0)
	>	101	Anita	6000.00	500.0	10
	>	102	Anish	5500.00	(NULL)	20
	>	103	Ankit	5200.00	300.0	20

5. What will be output if you are giving LIKE predicate as '%_%' ESCAPE '\'?

Answer:

```
-- LIKE '%\_%' ESCAPE '\'
```

```
-- Matches any string containing a literal underscore '_'
```

6. For each employee, display the employee number, job, salary, and salary increased by 15% and expressed as a whole number. Label the column New Salary.

Query:

```
SELECT emp_no, emp_name, emp_sal, ROUND(emp_sal * 1.15) AS "New Salary"
FROM Employee;
```

Output:

		emp_no decimal(3,0)	emp_name varchar(30)	emp_sal decimal(8,2)	New Salary
	>	101	Anita	6000.00	6900
	>	102	Anish	5500.00	6325
	>	103	Ankit	5200.00	5980
	>	104	Sunil	7000.00	8050
	>	105	Manoj	6500.00	7475

7. Modify your query no 4.(2) to add a column that subtracts the old salary from the new salary. Label the column Increase.

Query:

```
SELECT emp_no, emp_name, emp_sal, ROUND(emp_sal * 1.15) AS "New Salary",
ROUND(emp_sal * 0.15) AS "Increase" FROM Employee;
```

Output:

		emp_no decimal(3,0)	emp_name varchar(30)	emp_sal decimal(8,2)	New Salary	Increase
	>	101	Anita	6000.00	6900	900
	>	102	Anish	5500.00	6325	825
	>	103	Ankit	5200.00	5980	780
	>	104	Sunil	7000.00	8050	1050
	>	105	Manoj	6500.00	7475	975

8. Write a query that displays the employee's names with the first letter capitalized and all other letters lowercase, and the length of the names, for all employees whose name starts with J, A, or M. Give each column an appropriate label. Sort the results by the employees' last names.

Query:

```
SELECT CONCAT(UCASE(LEFT(emp_name, 1)), LCASE(SUBSTRING(emp_name, 2))) AS
"Proper Name", LENGTH(emp_name) AS "Name Length" FROM Employee WHERE
UPPER(SUBSTR(emp_name, 1, 1)) IN ('J', 'A', 'M') ORDER BY emp_name;
```

Output:

	Q	Proper Name	Name Length
>		Anish	5
>		Anita	5
>		Ankit	5
>		Manoj	5

9. Create a unique listing of all jobs that are in department 30. Include the location of the department in the output. (Use join)

Department

	Q	dept_no int(3)	dept_name varchar(30)	location varchar(30)
>		10	HR	NEW YORK
>		20	ACCOUNTING	CHICAGO
>		30	SALES	NEW YORK
>		40	RESEARCH	DALLAS
>		50	MARKETING	MUMBAI

Query:

```
SELECT DISTINCT j.job_title, d.location FROM Employee e JOIN Job j ON e.job_id =
j.job_id JOIN department d ON e.dept_no = d.dept_no WHERE e.dept_no = 30;
```

Output:

Q	job_title varchar	location varchar
>	Manager	NEW YORK
>	HR	NEW YORK

10. Write a query to display the employee name, department number, and department name for all employees who work in NEW YORK. (Use join)

Query:

```
SELECT e.emp_name, e.dept_no, d.dept_name FROM Employee e JOIN department d ON e.dept_no = d.dept_no WHERE d.location = 'New York';
```

Output:

Q	emp_name varchar	dept_no decimal	dept_name varchar
>	Anita	10	HR
>	Sunil	30	SALES
>	Manoj	30	SALES

11. Display the employee's last name and employee number along with their manager's last name and manager number. Label the columns Employee, Emp#, Manager, and Mgr#, respectively. (Use join)

Query:

```
SELECT e.emp_name AS Employee, e.emp_no AS "Emp#", m.emp_name AS Manager, m.emp_no AS "Mgr#" FROM Employee e JOIN Employee m ON e.manager_id = m.emp_no;
```

Output:

Q	Employee varchar	Emp# decimal	Manager varchar	Mgr# decimal
>	Anita	101	Sunil	104
>	Anish	102	Sunil	104
>	Ankit	103	Sunil	104
>	Manoj	105	Sunil	104

12. Write a query that displays the difference between the highest and lowest salaries. Label the column DIFFERENCE.

Query:

```
SELECT MAX(emp_sal) - MIN(emp_sal) AS DIFFERENCE FROM Employee;
```

Output:

Q	DIFFERENCE
>	1800.00

13. Create a query that will display the total number of employees and, of that total, the number of employees hired in 1995, 1996, 1997, and 1998.

Query:

```
SELECT COUNT(*) AS Total_Employees, SUM(YEAR(hire_date) = 1995) AS
Hired_1995, SUM(YEAR(hire_date) = 1996) AS Hired_1996, SUM(YEAR(hire_date) = 1997) AS
Hired_1997, SUM(YEAR(hire_date) = 1998) AS Hired_1998 FROM Employee;
```

Output:

	Total_Employees	Hired_1995	Hired_1996	Hired_1997	Hired_1998
>	5	1	1	1	2

14. Find the average salaries for each department without displaying the respective department numbers.

Query:

```
SELECT AVG(emp_sal) FROM Employee GROUP BY dept_no;
```

Output:

	AVG(emp_sal)
>	6000.000000
>	5350.000000
>	6750.000000

15. Write a query to display the total salary being paid to each job title, within each department.

Query:

```
SELECT d.dept_no, j.job_title, SUM(e.emp_sal) AS Total_Salary FROM Employee e
JOIN Job j ON e.job_id = j.job_id JOIN department d ON e.dept_no = d.dept_no GROUP BY
d.dept_no, j.job_title ORDER BY d.dept_name, j.job_title;
```

Output:

	dept_no int	job_title varchar	Total_Salary decimal
>	20	Analyst	5500.00
>	20	Salesman	5200.00
>	10	Clerk	6000.00
>	30	HR	6500.00
>	30	Manager	7000.00

16. Find the average salaries > 2000 for each department without displaying the respective department numbers.

Query:

```
SELECT AVG(emp_sal) FROM Employee WHERE emp_sal > 2000 GROUP BY dept_no;
```

Output:

	Q	AVG(emp_sal)	
>		6000.000000	
>		5350.000000	
>		6750.000000	

17. Give name of customers who are depositors having same branch city of mr. sunil. (Use sub query)

Query:

```
SELECT cname FROM deposit WHERE bname = (SELECT bname FROM deposit WHERE cname = 'Sunil');
```

Output:

	Q	cname	
>		Sunil	

18. Give deposit details and loan details of customer in same city where pramod is living. (Use sub query)

Query:

```
SELECT a_no AS id, cname, bname, amount, a_date AS transaction_date, 'Deposit' AS type FROM deposit WHERE bname = (SELECT bname FROM deposit WHERE cname = 'Pramod') UNION SELECT loanno AS id, cname, bname, amount, NULL AS transaction_date, 'Loan' AS type FROM borrow WHERE bname = (SELECT bname FROM deposit WHERE cname = 'Pramod');
```

Output:

Q	id	cname	bname	amount	transaction_date	type
	varchar	varchar	varchar	decimal	date	varchar
>	A03	Pramod	Delhi	3000.00	2006-07-30	Deposit
>	A05	Amit	Delhi	4500.00	2006-12-25	Deposit
>	L03	Pramod	Delhi	2500.00	(NULL)	Loan
>	L05	Amit	Delhi	4000.00	(NULL)	Loan

19. Create a query to display the employee numbers and last names of all employees who earn more than the average salary. Sort the results in ascending order of salary. (Use sub query)

Query:

```
SELECT emp_no, emp_name FROM Employee WHERE emp_sal > (SELECT  
AVG(emp_sal) FROM Employee) ORDER BY emp_sal;
```

Output:

	emp_no decimal	emp_name varchar
>	105	Manoj
>	104	Sunil

20. Give names of depositors having same living city as mr. anil and having deposit amount greater than 2000. (Use sub query)

Query:

```
SELECT cname FROM deposit WHERE bname = (SELECT bname FROM deposit  
WHERE cname = 'Anil') AND amount > 2000;
```

Output:

	cname varchar
>	Anil

21. Display the last name and salary of every employee who reports to ford. (Use sub query)

Query:

```
SELECT emp_name, emp_sal FROM Employee WHERE manager_id = (SELECT  
emp_no FROM Employee WHERE emp_name = 'Ford');
```

Output:

	emp_name varchar	emp_sal decimal
>		

22. Display the department number, name, and job for every employee in the accounting department. (Use sub query)

Query:

```
SELECT e.dept_no, e.emp_name, (SELECT j.job_title FROM Job j WHERE j.job_id =  
e.job_id) AS job_title FROM Employee e WHERE dept_no = (SELECT dept_no FROM  
department WHERE dept_name = 'ACCOUNTING');
```

Output:

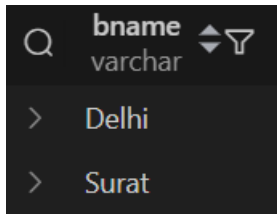
	dept_no decimal	emp_name varchar	job_title varchar
>	20	Anish	Analyst
>	20	Ankit	Salesman

23. List the name of branch having highest number of depositors. (Use sub query)

Query:

```
SELECT bname FROM deposit GROUP BY bname HAVING COUNT(cname) = (SELECT  
MAX(depositor_count) FROM (SELECT bname, COUNT(cname) AS depositor_count FROM  
deposit GROUP BY bname) AS sub);
```

Output:



bname varchar
> Delhi
> Surat